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||Jai Sri Gurudev||

Sri Adichunchanagiri Shikshana Trust (R)

SJB INSTITUTE OF TECHNOLOGY

(Affiliated to Visvesvaraya Technological University, Belagavi & Approved by AICTE, New Delhi)

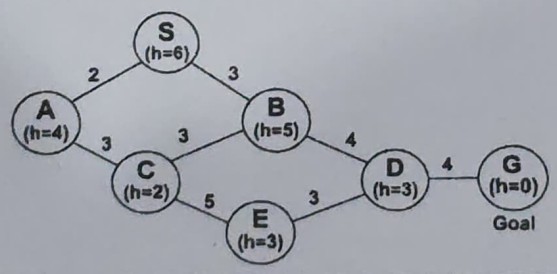
No. 67, BGS Health & Education City, Dr. Vishnuvardhan Road, Kengeri, Bengaluru-560060.

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

INTERNAL ASSESSMENT PAPER (VI Sem)

Internal Test: 1	Academic Year: ODD/2024-2025	
Subject : AI & ML	Sub-Code: 23CST601	Sem: VI
Date: 12/03/2026	Time: 9:00AM TO 10:30AM	Dur: 1.30 min
Internal Test max marks: **50**		
Staff-Incharge: **Mrs. Laxmi Shabadi / Mr. Santhosh K**		

I. Internal Test (Answer any two full questions choosing one from each part)

Main Ques	Sub Ques	Full Question	Marks	BT Level	CO'S
		Part A			
1.	A)	Distinguish between the following terms: i. Artificial Intelligence and Machine Learning ii. Problem space and state space iii. Blind search and Heuristic search	10	L4	CO1
	B)	Explain the concept of Problem Space in Artificial Intelligence. Represent the 8-puzzle problem as a state space with suitable diagram.	10	L2	CO1
	C)	Write the Steepest-Ascent Hill Climbing algorithm and explain its working.	5	L2	CO1
		OR			
2.	A)	Explain Heuristic Search with suitable example. Discuss the importance of heuristic functions in problem solving.	10	L2	CO1
	B)	Write the algorithm for A* search. Apply it to find the optimal path for a given graph when heuristic values are provided.  	10	L3	CO1
	C)	Define Artificial Intelligence. List and explain the characteristics of AI problems.	5	L1	CO1
		Part B			
3.	A)	Explain Univariate, Bivariate and Multivariate data analysis with suitable examples.	10	L2	CO2

	B)	Consider the following dataset: X Y 2 4 4 2 6 4 8 6 a) Compute the mean of X and Y. b) Construct the covariance matrix. c) Find eigenvalues and eigenvectors. d) Identify the first principal component using PCA.	10	L2	CO2
	C)	Define Version Space. Discuss the limitations of Find-S algorithm over candidate elimination algorithm.	5	L2	CO3
		OR			
4.	A)	Explain challenges in Machine Learning such as overfitting, data quality issues and scalability.	10	L2	CO2
	B)	Explain the Machine Learning pipeline / process with a neat diagram including data collection, preprocessing, model training and evaluation.	10	L2	CO2
	C)	Explain the four basic steps in the machine learning process?	5	L2	CO3

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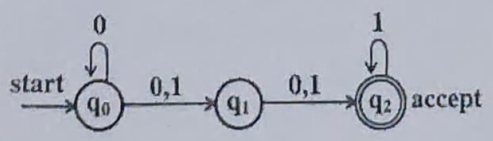
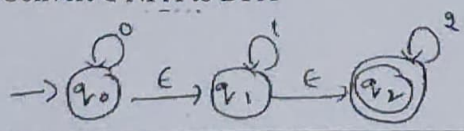
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

INTERNAL ASSESSMENT PAPER

Internal Test: I		Academic Year: EVEN 2025-26
Subject: Theory of Computation	Sub-Code: 23CSI602	Sem: VI
Date:12/03/2026	Time:2:30 to 4:00 pm	Dur:1.30 hour
Internal Test max marks: 50		
Staff-Incharge: Dr.Roopa M J ,Chetana,K N , Dr.Veena H N		

Internal Test (Answer any two full questions choosing one from each part)
(Each full question carries 25 marks)

Main Ques	Sub-Que	Full Question	Marks	BT level	CO
		Part A			
1	a	Define the following terms with example: i) Alphabet ii) Power of Alphabet iii) Languages iv) String v) Symbol	10	L1	CO1
	b	i. Obtain a DFA for language L={w: ending with ab|ba}where Σ={a,b} ii. Obtain a DFA to accept strings of a's and b's having odd number of a's and even number of b's.	10	L2	CO1
	c	List and explain applications of Finite Automata	5	L2	CO1
		OR			
2	a	Obtain a DFA for the following NFA 	10	L2	CO1
	b	Convert ϵ-NFA to DFA 	10	L2	CO1
	c	List the differences between DFA,NFA and ϵ-NFA	5	L2	CO1

Part B

3	a	Show that the following language is not regular where $L = \{ ww^R \mid w \in (0+1)^* \}$	10	L2	CO2
	b	Define Regular Expression. Write RE for the following languages: i) $L = \{a^n b^m \mid m+n \text{ is even}\}$ ii) $L = \{ a^n b^m \mid m \geq 2\ n \leq 4\}$ iii) $L = \{a^{2n} b^{2m} \mid n \geq 0, m \geq 0\}$	10	L2	CO2
	c	Obtain an NFA for the Regular expression (a*+b*+c*)	5	L2	CO2

OR

4	a	Show that the following language is not regular $L = \{ a^n b^n \mid n >= 0\}$	10	L2	CO2
	b	Minimize the following DFA	10	L2	CO2
	c	Show that Regular Languages are closed under complement and intersection.	5	L2	CO2

δ	0	1
→ A	B	F
B	G	C
* C	A	C
D	C	G
E	H	F
F	C	G
G	G	H
H	G	C

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

INTERNAL ASSESSMENT PAPER (VI Sem)

Internal Test: 1	Academic Year: EVEN/2025-2026	
Subject : Research Methodology & IPR	Sub-Code: 23RMAE61	Sem: VI
Date: 14/03/2026	Time: 2:30PM TO 4:00PM	Dur: 1.30 min
Internal Test max marks: **50**		
Staff-Incharge: **Dr. Naveena C / Dr. Bindiya M K**		

I. Internal Test (Answer any two full questions choosing one from each part)

Main Ques	Sub Ques	Full Question	Marks	BT Level	CO'S
		Part A			
1.	A)	Define the term research and explain the research flow cycle with a relevant diagram	10	L1	CO1
	B)	List the different types of research misconduct and provide a brief explanation for each one.	10	L1	CO1
	C)	Explain objectives of Engineering Research	5	L2	CO1
		OR			
2.	A)	With neat diagram explain the categories of knowledge in research	10	L2	CO1
	B)	Discuss the different types of engineering research. Clearly point out the differences between all of them with examples.	10	L2	CO1
	C)	What is hypothesis and Data analysis?	5	L1	CO1
		Part B			
3.	A)	Explain the ways in which new and existing knowledge contribute to the research process.	10	L2	CO2
	B)	Describe different Bibliography databases tools in detail	10	L2	CO2
	C)	Define Assets. Explain its different types give 5 examples for each.	5	L1	CO3
		OR			
4.	A)	Define IP. With a neat diagram explain different legal rights of IPR.	10	L2	CO3
	B)	Define citation and explain the different types of citation	10	L1	CO2
	C)	Define Acknowledgment and explain what should be included in .	5	L1	CO2

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An Autonomous Institute under VTU
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Department of Computer Science & Engineering

INTERNAL ASSESSMENT PAPER (6th sem)

Internal Test: I		Academic Year: ODD / 2025-26
Subject : Full Stack Development	Sub-Code: 23CSE644	Sem: 6th
Date:14/03/2026	Time: 8.45am - 10.15am	Dur: 1.30 min
Internal Test max marks: 50		
Staff-Incharge: Mrs Vijayalakshmi B/Mrs Shilpashree S		

I Internal Test (Answer any two full questions choosing one from each part)
(Each full question carries 25 marks)

Main Ques	Sub Ques	Full Question	Marks	BT Levels	CO Mapped
		Part A			
1.	A)	Explain the processing of Django request and response with a flow diagram.	10	L2	CO1
	B)	Illustrate the process of URL mapping with views.	10	L2	CO1
	C)	What are wild card patterns in URLs? Provide an example of how they are used in Django.	5	L2	CO1
		OR			
2.	A)	Define Web Framework? Explain with an example of design of a cWeb application written using the Common Gateway Interface (CGI) standard with its disadvantage.	10	L2	CO1
	B)	Write a python function in Django framework that displays date and time which is an offset of 4 hours from current date and time.	10	L3	CO1
	C)	Describe loose coupling in the context of Django URLConfs. Why is it important?	5	L2	CO1
		Part B			
3.	A)	Explain Basic Template Tags and Filters.	10	L2	CO3
	B)	Develop a simple Django app that displays an unordered list of fruits and an ordered list of selected students for an event.	10	L3	CO2
	C)	Illustrate the usage of the admin interface in Django.	5	L2	CO2
		OR			
4.	A)	Discuss template inheritance in Django with an example.	10	L2	CO3
	B)	Develop a django app that performs student registration to a course. It should also display a list of students registered for any selected course. Create student and course as models with enrollment as ManyToMany field.	10	L3	CO2
	C)	Explain Customizing the Admin Interface.	5	L2	CO2

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Recognized by UGC, New Delhi with 2(f) & 12 (B)
NBA
Department of Electrical and Electronics Engineering
(Accredited by NBA)

INTERNAL ASSESSMENT PAPER

Internal Assessment (Test): 1		Academic Year: 2025-26
Course Title: Alternate Energy Sources	Course Code: 23EEO611	Sem: 6th
Date: 13/03/26	Time: 2:30 – 4:00 pm	Duration: 1 Hrs 30 Min
Faculty In-charge: Vijay Kumar K		Max Marks:50

Note: Answer any two full questions, choosing one from each part. Each full question carries **25 marks**.

Q No.	Question	Marks	RBT Levels	CO
	Part – A			
1.a	Explain the causes of energy scarcity. What is the solution to energy scarcity?	9	L2	CO1
1.b	With the help of diagram define a) Solar altitude angle b) hour angle c) Solar azimuth angle d) Declination angle e) equation of time	10	L2	CO2
1. c	Explain basic Rankine cycle with a neat diagram	6	L2	CO2
	OR			
2.a	With a neat diagram explain the layers of Sun	10	L2	CO2
2.b	Calculate zenith angle of the sun at Lucknow (26.75^0 N) at 9:30 am on February 16, 2012.	6	L3	CO2
2.c	With a neat block diagram explain the Solar Thermal Energy Applications	9	L2	CO1
	Part – B			
3.a	With neat diagram explain solar pond and write any two advantages of it.	9	L2	CO3
3.b	Discuss the applications, adavantage and disadvantage of hydrogen storage	10	L2	CO4
3.c	With a neat diagram explain Parabolic Dish thermal collector system	6	L2	CO3
	OR			
4.a	With a neat diagram explain any one type of geothermal based electric power generation	10	L2	CO4
4.b	Explain with neat sketches i) flat plate air collectors and ii) flat plate liquid collectors	8	L2	CO3
4.c	With the help of neat diagram, explain key elements of photoVoltaic cell	7	L2	CO3